



Analysis of energy usage in Nd–Fe–B magnet to magnet recycling

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Highlights

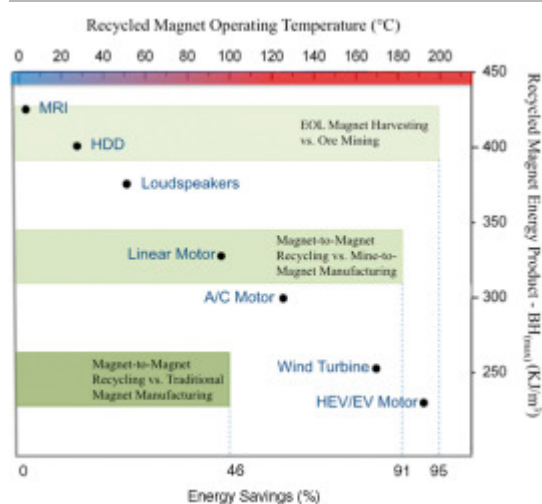
- Environmental and economic benefits of magnet-to-magnet recycling.
- Restoration of magnetic properties.
- High temperature demagnetization loops for recycled magnet.
- Recycled magnets yield higher coercivity value than the starting magnets.

Abstract

The recycling of Nd–Fe–B-type sintered magnets has received a great deal of attention over the past 15 years, but commercially viable and environmentally friendly solutions have remained elusive, until now. Here we report on the recycling of waste magnets using a novel “magnet-to-magnet” approach and show that it can work on a commercial scale. As part of the study, we have evaluated material supply, energy consumption, and

the overall environmental and economic benefits of magnet-to-magnet recycling, compared to the traditional manufacturing of “virgin” magnets. We also demonstrate how the magnet-to-magnet recycling approach can reduce energy consumption by more than 45%, during magnet production, with overall savings exceeding 90% when we also consider raw-materials production.

Graphical abstract



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Introduction

Of the commercially useful metals, rare-earth elements (REEs) are among the most critical in terms of supply. The REE group is made up of 17 elements (the lanthanides along with Sc and Y) that are usually found together in minerals such as bastnäsite, monazite, xenotime and in ion-adsorbed clays. Given the high demand and the strained supply of REEs for magnetism, lighting, catalysis, and displays for computers and TV equipment (leading to significant recent price volatility Hatch, 2014a), various reports released by UNEP (Buchert et al., 2009), the European Commission (Critical Raw Materials for the EU, 2010, Defining Critical Raw Materials, 2012), and the USGS (US Geological Survey, 2010), among others (Jones, 2011, Hatch, 2012, British Geological Survey, 2011), highlight the need to look at a wide range of options to ensure the secure supply of REEs. While new REE deposits have been identified (Hatch, 2014b), the construction of new mines is associated with high capital costs and timelines of more than 10–15 years before they can start to operate. Thus, the recycling of REEs has the potential to be far more economical and more readily achievable than the exploitation of new mineral deposits.

REEs have specific physical properties that make them ideal for inclusion in permanent-magnet alloys. In particular, their high spin–orbit coupling is exploited to create magnetocrystalline anisotropy, which leads to high values of coercivity (Skomski and Coey, 1999). Nd–Fe–B-type magnets contain a magnetically hard phase based on (Nd, Pr, Dy)–Fe–B and other trace elements, with REE contents varying from 27 to 32 wt%, Fe ranging from 67 to 73 wt%, B at 1 wt% (Sagawa et al., 1984, Peiró et al., 2013), and other minor additions of transition metals. Sm–Co magnets were the first commercial REE magnets; however, but are now generally limited to use in high-temperature applications, where the lower Curie temperature (T_c) of sintered Nd–Fe–B magnets can be problematic (Sugimoto, 2011).

Various recycling approaches for the recovery of sintered Nd–Fe–B magnets have been suggested (as described in detail in the Related Work section). These include waste-to-alloy, hydro-metallurgical, pyro-metallurgical, melting, gas-phase extraction, and magnet-to-magnet approaches. In the present work, we concentrate on the magnet-to-magnet route, which was recently shown to work on a commercial scale (Zakotnik and Tudor, 2014). This technique has a major advantage, not only because of its simplicity, but also because of its lower cost when compared to traditional recycling methods. The technique has been significantly improved and scaled-up, and shows excellent results on a commercial scale, with improved magnetic performance, as will be described here. In comparison with other recycling techniques, the magnet-to-magnet approach allows for the 10 or so metals present in a modern sintered Nd–Fe–B magnets to be recycled simultaneously. Moreover, it allows for the development of a closed-loop recycling system for REE-based magnets, thus considerably reducing the emissions generated by the extraction and processing of the REEs and transition metals present in these magnets.

One important step in magnet-to-magnet recycling is the separation of the magnets from the products in which they are embedded. Nd–Fe–B magnets are generally assembled into intermediate products (e.g., electric motors and voice-coil motors (VCMs)), which are part of end products, such as conventional automobiles, electric vehicles (EVs), wind-turbine generators, magnetic-resonance-imaging (MRI) machines, electronic devices (e.g., earphones and speakers), hard-disk drives (HDDs), as well as more challenging future applications like the magnetic-levitation (‘mag-lev’) trains. It is estimated that the global market breakdown of Nd–Fe–B magnets is 35% for computers, 25% for audio systems, 15% for wind turbines, 15% for automobiles, 5% for household appliances, and 5% for MRI machines (Du and Graedel, 2011). As advanced technologies are projected to increase significantly, so is the demand for REEs (Rieger, 2014). Current market predictions for future energy demands indicate that the demand for REE permanent magnets will continue to see a strong and steady growth.

While the Nd–Fe–B magnets contained in wind turbines or MRI machines are easily extracted for recycling, the Nd–Fe–B magnets contained in consumer goods are not yet successfully separated, because no regulation is yet enforced for their recycling. In fact, it is safe to assume that almost all Nd–Fe–B magnets coming from this latter group are directly shredded together with the end product. The amount and quality of the REEs in the resulting material mixture are very low and, thus, their recovery may require a long sequence of separation processes. To alleviate this problem, the recycling processes for Nd–Fe–B magnets must be simple and be able to recover the quality of the metals in order to ensure good standards for input back into the market.

In this work, we compare the traditional method of manufacturing sintered Nd–Fe–B magnets with the magnet-to-magnet approach used to recycle these magnets. The savings in energy consumption when using a 1000-ton manufacturing plant are illustrated in the Results section. We conclude that the availability of rare-earth metals is fundamental to the development of current and future technologies that use sintered Nd–Fe–B magnets, helping society on its road to a low-carbon economy.

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Section snippets

Related work

Schüler et al. (2011) and Binnemans et al. (2013) have described various recycling approaches for the recovery of sintered Nd–Fe–B magnets. These include waste-to-alloy, hydro-metallurgical, pyro-metallurgical, melting, gas-phase extraction, and magnet-to-magnet approaches. Most recycling solutions to date are restricted to laboratory scale and operate only under idealized conditions. In addition, the properties of the resulting magnets/powders from these studies show reduced magnetic ...

Production of Nd–Fe–B magnets

Fig.1 contrasts the two approaches for manufacturing sintered Nd–Fe–B magnets: (1) the traditional route, highlighted in blue and gray, and (2) the magnet-to-magnet recycling

route, highlighted in orange and gray. ...

Magnetic properties of recycled magnets

To the best of our knowledge, the magnet-to-magnet recycling approach referenced here (Zakotnik and Tudor, 2014) is different to any other recycling approach. It is also the only approach that has been tested in mass production. This section provides the reader with a selection of the results obtained from recycling about 5 tons of waste sintered magnets. We demonstrate that mixing the waste magnets with carefully prepared Nd–Pr additives results in not just total recovery, but also in an ...

Results of energy consumption

The prices of REEs and sintered magnets are strongly affected by the costs for the energy, transportation and labor required. In this section, we will focus on the energy consumption, comparing the two routes, traditional and recycling, for manufacturing sintered Nd–Fe–B magnets. The results are shown in Table 5, where we list the processing steps needed for both routes, together with their corresponding equipment characteristics and annual energy consumptions.

Our case study focuses on ...

Conclusions

In this study, we have compared the energy usage of magnet to magnet recycling with that of the traditional mining and strip casting method. A commercial study of producing Nd–Fe–B magnets from recycled waste magnetic material coming from end-of-life applications (Zakotnik and Tudor, 2014) has shown that this process is economically, environmentally, and energetically favorable. Five tons of waste sintered magnets were collected from consumer products, and new magnets were produced with ...

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